

1. The system impulse response is  $h[n] = \delta[n+1] - 2\delta[n] + 2\delta[n-1] + \delta[n-2]$ .
  - (a) Let  $x[n] = \delta[n] + 2\delta[n-1] - \delta[n-2]$ . Find  $x[n] * h[n]$ . (6%)
  - (b) Find the transform function  $H(z)$  and ROC. (6%)
  - (c) Find the difference equation for input  $x[n]$  and output  $y[n]$ . (5%)
2. Consider a causal LTI system with input  $x[n]$  and output  $y[n]$ , and the system function  $H(z) = (30 - 3z^{-1} + 2z^{-2}) / (10 - 3z^{-1} - z^{-2})$ .
  - (a) Find system impulse response  $h[n]$ . (7%)
  - (b) Find the difference equation for input  $x[n]$  and output  $y[n]$ . (5%)
  - (c) Is the system a stable system? (5%)
  - (d) Determine  $h[0]$  using the initial-value theorem. (5%)
3. Please decide  $L$ ,  $M$  and  $\omega_c$  in Fig. 1 with the respective conditions of (a) and (b).
  - (a) The sampling frequencies of  $x_1[n]$  and  $x_2[n]$  are 20kHz and 14kHz respectively. (6%)
  - (b) The sampling frequencies of  $x_1[n]$  and  $x_2[n]$  are 12kHz and 18kHz respectively. (6%)
  - (c) What is the maximum bandwidth of  $x_1[n]$  that will avoid signal distortion when  $M=5$  and  $L=3$  in Fig. 1? (6%)

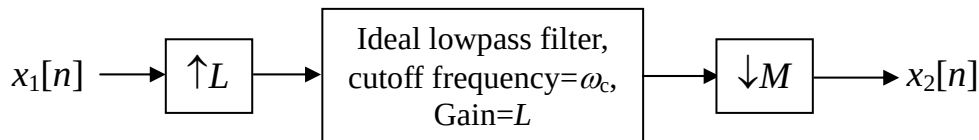


Fig. 1.

4. Let  $X(e^{j\omega})$  be the DTFT of  $x[n]$ .
  - (a) Find the inverse Fourier transform of  $X(e^{-j\omega})$ . (6%)
  - (b) Find the DTFT of  $x^*[n]$ . (6%)
5. Let  $x[n]$  and  $h[n]$  denote complex sequences and  $X(e^{j\omega})$  and  $H(e^{j\omega})$  be their respective Fourier transforms. Show that the Fourier transform of  $x[n] * h[n]$  is  $X(e^{j\omega})H(e^{j\omega})$ . (7%)
6. Assume that  $x[n]$  is real and even; i.e.  $x[n] = x[-n]$ . Further, assume that  $z_0$  is a zero of  $X(z)$ ; i.e.  $X(z_0) = 0$ .
  - (a) Show that  $1/z_0$  is also a zero of  $X(z)$ . (7%)
  - (b) Are there other zeros of  $X(z)$  implied by the information given? (7%)

7. In the system of Fig. 2a, where  $H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \pi/L \\ 0, & \pi/L \leq |\omega| \leq \pi \end{cases}$ .

Sketch  $Y_c(j\Omega)$  if  $X_c(j\Omega)$  is as shown in Fig. 2b. (10%)

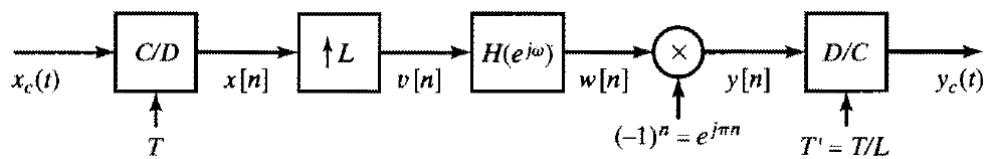


Fig. 2a.

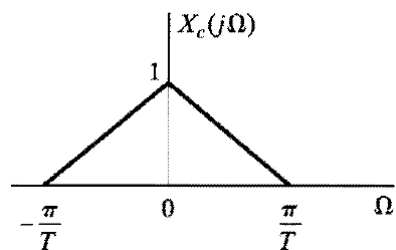


Fig. 2b